

Networking & TCP/IP Teaching Sequence

1. Layered Thinking and Encapsulation

Focus on understanding how data is encapsulated as it moves from application to wire.

Emphasize why failures propagate upward and why packet captures show multiple protocol layers simultaneously.

Certification alignment: Network+ domain 1.1, Security+ domain 2.4.

2. Layer 2 vs Layer 3 Mechanics

MAC vs IP responsibility and how ARP bridges the two.

Why local attacks dominate internal networks.

Certification alignment: Network+ domain 1.5, Security+ domain 3.1.

3. Routing and Trust Boundaries

How routers make forwarding decisions and where segmentation truly exists.

Impact of default routes and misrouting.

Certification alignment: Network+ domain 2.3.

4. Transport Layer Behavior

TCP statefulness vs UDP statelessness.

Why reliability assumptions fail under attack.

Certification alignment: Network+ domain 1.3, Security+ domain 3.2.

5. DNS as Infrastructure

DNS resolution path and caching.

Why DNS is both a dependency and an attack surface.

Certification alignment: Network+ domain 1.2, Security+ domain 2.2.

6. HTTP(S) and Application State

Stateless protocols layered over stateful sessions.

Limits of TLS protection.

Certification alignment: Security+ domain 1.4.

7. ICMP and Network Feedback

Role of ICMP in diagnostics and MTU discovery.

Operational risk of blocking ICMP.

Certification alignment: Network+ domain 1.6.